

ORIGINAL RESEARCH

National Impact of Optimal Implementation of Guideline-Directed Medical Therapy in Heart Failure With Reduced Ejection Fraction

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ABSTRACT

BACKGROUND Despite robust evidence that quadruple guideline-directed medical therapy (GDMT), comprising angiotensin receptor-neprilysin inhibitors (ARNIs), evidence-based beta-blockers, mineralocorticoid receptor antagonists (MRAs), and sodium-glucose cotransporter 2 (SGLT2) inhibitors, reduces mortality and hospitalizations in heart failure with reduced ejection fraction (HFrEF), implementation in clinical practice remains markedly incomplete.

OBJECTIVES This study aimed to provide updated national estimates of the eligible untreated HFrEF population and to quantify deaths and hospitalizations preventable with optimal implementation of quadruple GDMT in the United States.

METHODS The authors performed a population-level decision analytic modeling study using contemporary U.S. epidemiologic data. National HFrEF prevalence estimates were derived from the American Heart Association Heart Disease and Stroke Statistics 2026 report, and annual HFrEF hospitalization counts were derived from the National Inpatient Sample 2022-2023. Eligible untreated populations were estimated after sequential exclusions and therapy-specific contraindication adjustments using primary treatment rates from Epic Cosmos 2023-2025, with sensitivity analyses using alternative treatment-rate sources. Trial-derived numbers needed to treat and relative risk reductions were applied to estimate deaths preventable over 12 months and annual heart failure (HF) hospitalizations prevented.

RESULTS An estimated 2.76 million U.S. adults with chronic symptomatic HFrEF were eligible for quadruple GDMT, yet only 18.2% received it. Eligible untreated populations included 733,891 for beta-blockers, 1,856,531 for ARNIs, 1,543,967 for MRAs, and 1,717,444 for SGLT2 inhibitors. Class-specific projected deaths preventable over 12 months were 26,210 for beta-blockers, 34,495 for ARNIs, 26,620 for MRAs, and 26,422 for SGLT2 inhibitors; the aggregate estimate across treatment gaps was 113,747 (95% uncertainty interval [UI]: 90,173-149,875). Optimal implementation was also projected to prevent 357,332 HF hospitalizations annually (95% UI: 297,493-425,674).

CONCLUSIONS Incomplete implementation of quadruple GDMT in HFrEF remains a major modifiable opportunity to reduce preventable deaths and HF hospitalizations in the United States. (JACC Heart Fail. 2026;■:103295)
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**ABBREVIATIONS
AND ACRONYMS****ACEI** = angiotensin-converting enzyme inhibitor**ARB** = angiotensin receptor blocker**ARNI** = angiotensin receptor-neprilysin inhibitor**GDMT** = guideline-directed medical therapy**HF** = heart failure**HFrEF** = heart failure with reduced ejection fraction**LVEF** = left ventricular ejection fraction**MRA** = mineralocorticoid receptor antagonist**NNT** = number needed to treat**RASi** = renin-angiotensin system inhibitor**RRR** = relative risk reduction**SGLT2** = sodium-glucose cotransporter 2

Hear failure (HF) represents a major and growing public health burden in the United States.¹

Contemporary U.S. national estimates indicate that 7.7 million Americans aged 20 years or older are living with HF, up from 6.7 million in 2017–2020,¹ with prevalence projected to reach 8.7 million by 2030 and 11.4 million by 2050.² HF remains a major contributor to mortality, serving as a contributing cause in 425,147 deaths in 2022 and accounting for 45% of cardiovascular deaths.² Beyond mortality, HF accounts for approximately 1.2 million primary hospitalizations annually, with heart failure with reduced ejection fraction (HFrEF) comprising roughly half of these events and therefore representing a key target for therapies shown in randomized trials to reduce hospitalization and improve survival.²

Randomized trials have established that renin-angiotensin system inhibition (angiotensin-converting enzyme inhibitor [ACEI]/angiotensin receptor blocker [ARB]),^{3,4} angiotensin receptor-neprilysin inhibition (ARNI),⁵ evidence-based beta-blockers,^{6,7} mineralocorticoid receptor antagonists (MRAs),⁸ and sodium-glucose cotransporter 2 (SGLT2) inhibitors^{9,10} each meaningfully reduce mortality and HF hospitalization among patients with HFrEF. Based on this evidence, contemporary professional guidelines strongly recommend initiation of a foundational quadruple regimen, ARNI (or ACEI/ARB), an evidence-based beta-blocker, an MRA, and an SGLT2 inhibitor, for patients with chronic HFrEF and left ventricular ejection fraction (LVEF) 40% or less, in the absence of contraindications.

Despite strong evidence, implementation of guideline-directed medical therapy (GDMT) remains incomplete, with substantial gaps in adoption across health systems.¹¹ Contemporary data suggest that only a minority of eligible patients receive all 4

foundational classes, with recent U.S. estimates of quadruple therapy use remaining below 20%.^{11,12} Registry and claims-based studies further illustrate persistent shortfalls at the class level.^{13,14} This large eligible but undertreated population translates into substantial preventable morbidity and mortality. Updated national estimates of deaths and hospitalizations that could be averted through optimal quadruple GDMT implementation are therefore essential for evidence-based policymaking, benchmarking health care system performance, and appraising the population health impact of improving HFrEF care relative to other competing clinical and public health priorities.

In this context, using national prevalence estimates and hospitalization counts, we modeled the U.S. population with chronic HFrEF eligible for each GDMT class after applying standard exclusions and contraindication adjustments, quantified the eligible untreated population under contemporary treatment patterns, and applied trial-based numbers needed to treat (NNTs) and relative risk reductions (RRRs) to estimate projected deaths preventable over 12 months, annual HF hospitalizations preventable as the primary hospitalization outcome, and all-cause hospitalizations preventable as supportive secondary estimates with optimal implementation of individual therapy gaps and total quadruple GDMT implementation. These contemporary estimates are intended to inform evidence-based policymaking and appraisal of the population-level benefits of improving GDMT implementation.

METHODS

STUDY DESIGN AND ANALYTICAL FRAMEWORK. We performed a population-level decision analytic modeling study to estimate the potential reduction in mortality and hospitalizations associated with optimal implementation of quadruple GDMT among patients with chronic HFrEF in the United States. The analytic framework extended previous GDMT

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

implementation models by incorporating updated national epidemiologic estimates, contemporary treatment patterns, therapy-specific contraindications, and current hospitalization data. The analysis sequentially estimated: 1) the U.S. population with chronic symptomatic HFrEF eligible for GDMT; 2) the eligible untreated population under contemporary treatment patterns; 3) projected deaths preventable over 12 months using trial-derived NNT values standardized to a 12-month horizon; and 4) annual HF hospitalizations prevented as the primary hospitalization outcome and all-cause hospitalizations prevented as supportive secondary estimates using randomized trial-derived RRRs. Modeled estimates are reported with 95% uncertainty intervals (UIs) derived from probabilistic sensitivity analysis. The cohort derivation and sequential exclusions are summarized in [Figure 1](#).

DATA SOURCES AND MODEL INPUTS. Model inputs and data sources are summarized in [Supplemental Table 1](#), with additional data-source definitions provided in the [Supplemental Methods](#). National HF prevalence estimates were obtained from the American Heart Association Heart Disease and Stroke Statistics 2026 report using National Health and Nutrition Examination Survey 2021-2023 data.¹ The proportion of patients with LVEF $\leq 40\%$ was derived from the GWTG-HF (Get With The Guidelines-Heart Failure) registry (2014-2021).^{15,16} Estimates of the proportions of patients with NYHA functional class I symptoms, hospice or comfort-care status, and receipt of advanced therapies were derived from the CHAMP-HF (Change the Management of Patients With Heart Failure) registry (2015-2017)^{13,17} and National Health and Nutrition Examination Survey data (2015-2018)¹⁸ and were applied as exclusion criteria.

Annual primary and total HF hospitalization counts were obtained from the National Inpatient Sample 2022-2023. The proportion of HF hospitalizations attributable to HFrEF was applied to estimate HFrEF-attributable hospitalizations. To approximate hospitalizations among patients with established chronic HFrEF, we conservatively excluded 20% of events assumed to represent new-onset HF hospitalizations, avoiding attribution of preventability to index presentations of previously undiagnosed HF. This estimate was selected as a conservative literature-informed midpoint because reported proportions of de novo or recently diagnosed HF vary by data source, population, lookback period, and definition.^{19,20}

Contraindication rates were derived from GWTG-HF 2016-2023¹¹ and reflected practical

medication-specific ineligibility criteria based on regulatory labeling, guideline recommendations, and expert consensus, as detailed in the [Supplemental Methods](#). These rates were applied to estimate therapy-specific eligibility. Contemporary treatment rates for the primary analysis were derived from Epic Cosmos 2023-2025,¹² with available prescription rates for beta-blockers, renin-angiotensin system inhibitors (RASIs), ACEI, ARB, ACEI/ARB, ARNI, MRA, and SGLT2 inhibitor. Alternative treatment-rate estimates from GWTG-HF 2021-2024, GWTG-HF 2016-2023,¹¹ and Optum electronic health records 2020-2023²¹ were retained for sensitivity analyses.

DERIVATION OF GDMT ELIGIBILITY AND CONTEMPORARY TREATMENT GAPS.

The total U.S. HFrEF population was estimated by applying the proportion with LVEF $\leq 40\%$ to national HF prevalence estimates. To approximate chronic HFrEF eligible for long-term GDMT, we sequentially excluded patients with NYHA functional class I symptoms, hospice or comfort-care status, and advanced therapies. Therapy-specific eligibility was then determined by applying contraindication rates to this chronic HFrEF NYHA functional class II-IV cohort. Eligible untreated patients were estimated by applying contemporary treatment gaps to therapy-eligible populations ([Tables 1 and 2](#)).^{11,12,21}

For ARNI, the eligible untreated population was decomposed into patients receiving ACEI/ARB but not ARNI and patients receiving no RASi. Because ACEI/ARB and ARNI prescription categories in Epic Cosmos¹² were not mutually exclusive within the reporting window, mutually exclusive ARNI implementation groups were derived using any RASi and ARNI rates: ACEI/ARB without ARNI was calculated as any RASi use minus ARNI use, and no RASi as 1 minus any RASi use.

MORTALITY MODELING AND ESTIMATION OF DEATHS PREVENTABLE OVER 12 MONTHS.

RRRs and NNT values for all-cause mortality were obtained from pivotal randomized trials of GDMT in HFrEF ([Supplemental Table 2](#)). Trial-reported NNTs were standardized to a 12-month horizon using a time-adjusted absolute risk reduction approach, assuming approximately constant hazards over follow-up; the formula is provided in the [Supplemental Methods](#).²²⁻²⁴

For ARNI, mortality benefit was modeled separately for patients switching from ACEI/ARB to ARNI and for patients initiating ARNI from no RASi. For ACEI/ARB-to-ARNI switching, we used the direct PARADIGM-HF (Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure)⁵ ARNI vs ACEI

TABLE 1 Projected U.S. HFrEF Population, Hospitalization Burden, and Therapy Eligibility for GDMT

U.S. Value (Point)	
HF prevalence	7,700,000 (6,468,000-9,548,000)
HFrEF prevalence	3,280,200 (2,613,882-4,250,104)
HFrEF NYHA functional class II-IV after exclusions	2,758,648 (2,197,124-3,578,600)
Primary HF hospitalizations	1,272,855
Total HF hospitalizations	6,038,935
Primary HFrEF hospitalizations	584,240 (501,801-666,379)
Total HFrEF hospitalizations	2,771,871 (2,380,747-3,161,571)
Primary HFrEF hospitalizations after exclusion	467,392 (372,904-550,133)
Total HFrEF hospitalizations after exclusion	2,217,497 (1,769,207-2,610,051)
Treatment rates, %	
Beta-blocker	73.1
ACEI/ARB/ARNI (any RASi)	67.3
ARNI	26.7
MRA	35.5
SGLT2 inhibitor	33.6
Quadruple GDMT	18.2
Contraindications, %	
Beta-blocker	1.1
ARNI	8.2
MRA	13.2
SGLT2 inhibitor	6.3
Patients eligible for GDMT classes	
Beta-blocker	2,728,303 (2,173,001-3,539,158)
ARNI	2,532,439 (2,016,055-3,286,790)
MRA	2,394,507 (1,904,180-3,107,163)
SGLT2 inhibitor	2,584,853 (2,058,126-3,353,865)

Values are n (95% UI) or n. Model inputs and projected U.S. HFrEF NYHA functional class II-IV population used to estimate eligibility for quadruple GDMT and hospitalization burden. Data sources for model inputs are listed in [Supplemental Table 1](#). HF prevalence was obtained from AHA Heart Disease and Stroke Statistics; the interval shown reflects the published 95% CI. All other 95% UIs for modeled population and hospitalization estimates were derived using probabilistic sensitivity analysis. The HFrEF population was estimated by applying the proportion with LVEF $\leq$ 40% and excluding NYHA functional class I, hospice/comfort care, and advanced therapies. HF hospitalization counts were obtained from the National Inpatient Sample; HFrEF-attributable hospitalizations were estimated using a 45.9% HFrEF attribution proportion and a conservative 20% exclusion for new-onset HF. Contemporary treatment rates were derived from Epic Cosmos 2023-2025. Medication-specific contraindication rates were derived from GWTG-HF and applied separately to estimate therapy-specific eligibility. Quadruple GDMT was defined descriptively as any RASi, beta-blocker, MRA, and SGLT2 inhibitor.

ACEI = angiotensin-converting enzyme inhibitor; AHA = American Heart Association; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; GDMT = guideline-directed medical therapy; GWTG-HF = Get With The Guidelines-Heart Failure; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; RASi = renin-angiotensin system inhibitor; SGLT2 = sodium-glucose cotransporter 2; UI = uncertainty interval.

TABLE 2 Untreated Eligible Patients and Projected Deaths Preventable Over 12 Months With Optimal Implementation of Quadruple GDMT

Point	
Eligible patients not receiving GDMT	
Beta-blocker	733,891 (583,895-952,620)
ARNI: ACEI/ARB to ARNI	1,027,979 (817,172-1,333,971)
ARNI: no RASi to ARNI	828,552 (658,739-1,075,175)
ARNI: total	1,856,531 (1,475,815-2,409,233)
MRA	1,543,967 (1,226,502-2,005,823)
SGLT2 inhibitor	1,717,444 (1,365,982-2,229,938)
Projected deaths preventable over 12 mo	
Beta-blocker	26,210 (20,097-35,694)
ARNI: ACEI/ARB to ARNI	12,691 (9,745-17,260)
ARNI: no RASi to ARNI	21,804 (16,715-29,694)
ARNI: total	34,495 (26,977-46,150)
MRA	26,620 (20,408-36,251)
SGLT2 inhibitor	26,422 (20,233-35,949)
Total quadruple GDMT implementation (aggregate across treatment gaps)	113,747 (90,173-149,875)

Values are n (95% UI). Projected eligible untreated counts and deaths preventable over 12 months with optimal implementation of quadruple GDMT among U.S. patients with chronic HFrEF and NYHA functional class II-IV symptoms. Eligible untreated counts reflect therapy-specific eligible populations after applying contemporary Epic Cosmos treatment rates. For ARNI, benefits were decomposed into patients switching from ACEI/ARB to ARNI, using the PARADIGM-HF ARNI vs ACEI NNT, and patients initiating ARNI from no RASi, using the ARNI vs imputed placebo NNT. Projected deaths preventable over 12 months were calculated by dividing eligible untreated counts by the corresponding 12-month standardized NNT for all-cause mortality. The total quadruple GDMT implementation estimate represents aggregate projected deaths preventable across therapy-specific treatment gaps and should not be interpreted as a count of unique individuals. The 95% UIs were derived using probabilistic sensitivity analysis.

NNT = number needed to treat; PARADIGM-HF = Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure; other abbreviations as in [Table 1](#).

and should not be interpreted as mutually exclusive individuals ([Table 2](#), [Figure 2](#)).²⁵ Real-world effectiveness sensitivity analyses for mortality are presented in [Supplemental Tables 3 to 5](#).²⁶

mortality NNT. For no-RASi-to-ARNI initiation, we used an ARNI vs imputed placebo estimate derived by combining PARADIGM-HF⁵ and CHARM-Alternative⁴ (Candesartan in Heart Failure Assessment of Reduction in Mortality and Morbidity - Alternative) treatment effects on the log scale, with uncertainty propagation described in the [Supplemental Methods](#).

Projected deaths preventable over 12 months were calculated by dividing eligible untreated counts by 12-month standardized NNTs. The total quadruple GDMT implementation estimate represents aggregate projected deaths preventable across therapy-specific implementation gaps, assumes additive mortality benefits across therapies consistent with previous comparative and network meta-analytic analyses,

ESTIMATION OF PREVENTED HOSPITALIZATIONS. National primary and total HF hospitalizations were converted to chronic HFrEF hospitalization counts by applying the HFrEF hospitalization proportion and excluding presumed new-onset HF hospitalizations. HF hospitalizations were considered the primary hospitalization outcome because they are more directly attributable to GDMT effects; all-cause hospitalization estimates were supportive secondary estimates. Class-specific hospitalization effects were obtained from randomized trials and expressed as RRRs, with ARNI and SGLT2 inhibitor effect derivations described in the [Supplemental Methods](#) and summarized in [Supplemental Table 6](#). Baseline annual event rates were

calculated as annual chronic HFrEF hospitalizations divided by the modeled chronic HFrEF NYHA functional class II-IV population. Prevented hospitalizations were estimated as eligible untreated patients $\times$ baseline event rate $\times$ trial-derived RRRs (Supplemental Methods).

For total quadruple GDMT implementation, projected hospitalization benefits were calculated as the sum of projected hospitalizations prevented by closing individual beta-blocker, ARNI, MRA, and SGLT2 inhibitor treatment gaps, rather than by applying a separate quadruple-therapy uptake rate. Projected annual HF and all-cause hospitalizations prevented are presented in Table 3. HF hospitalizations are emphasized in Figure 3, whereas combined HF and all-cause hospitalization estimates are shown in Supplemental Figure 1 as supportive secondary estimates.

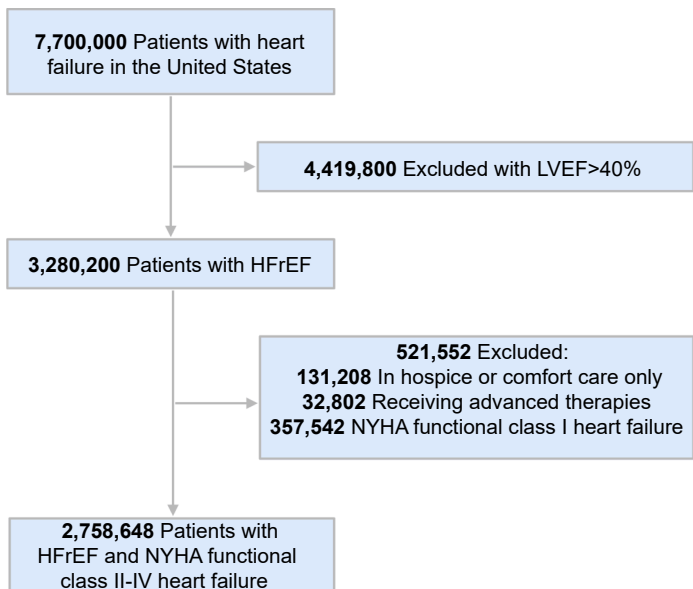
SENSITIVITY ANALYSES. Sensitivity analyses used alternative treatment-rate estimates from GWTG-HF 2021-2024, GWTG-HF 2016-2023,¹¹ and Optum EHR 2020-2023.²¹ Mortality and hospitalization calculations were repeated using each source-specific treatment-rate assumption. We also applied 80% and 60% of trial-derived treatment benefit to mortality and hospitalization projections to evaluate more conservative real-world effectiveness scenarios. Sensitivity analyses are presented in Supplemental Tables 3 to 5 and 7 to 9.

STATISTICAL ANALYSIS. Fixed-effect inverse-variance meta-analyses were performed in Stata version 18 (StataCorp). Population-level modeling and probabilistic sensitivity analyses were conducted using Python version 3.13.2. For modeled population estimates, deaths preventable over 12 months, and hospitalizations prevented, uncertainty was propagated using probabilistic sensitivity analysis across prespecified parameter distributions described in the Supplemental Methods. Point estimates were calculated from base-case inputs, and 95% UIs were derived from the 2.5 and 97.5 percentiles of simulated results. Trial-derived treatment-effect inputs are reported with 95% CIs, whereas modeled outputs are reported with 95% UIs (Supplemental Methods).

RESULTS

MODELED U.S. HFrEF POPULATION AND GDMT ELIGIBILITY. Using contemporary national epidemiologic estimates, approximately 7.7 million adults in the United States were living with HF, corresponding to an estimated 3,280,200 individuals with HFrEF after applying the proportion with LVEF $\leq 40\%$.

FIGURE 1 Flowchart for the HFrEF NYHA Functional Class II-IV Cohort

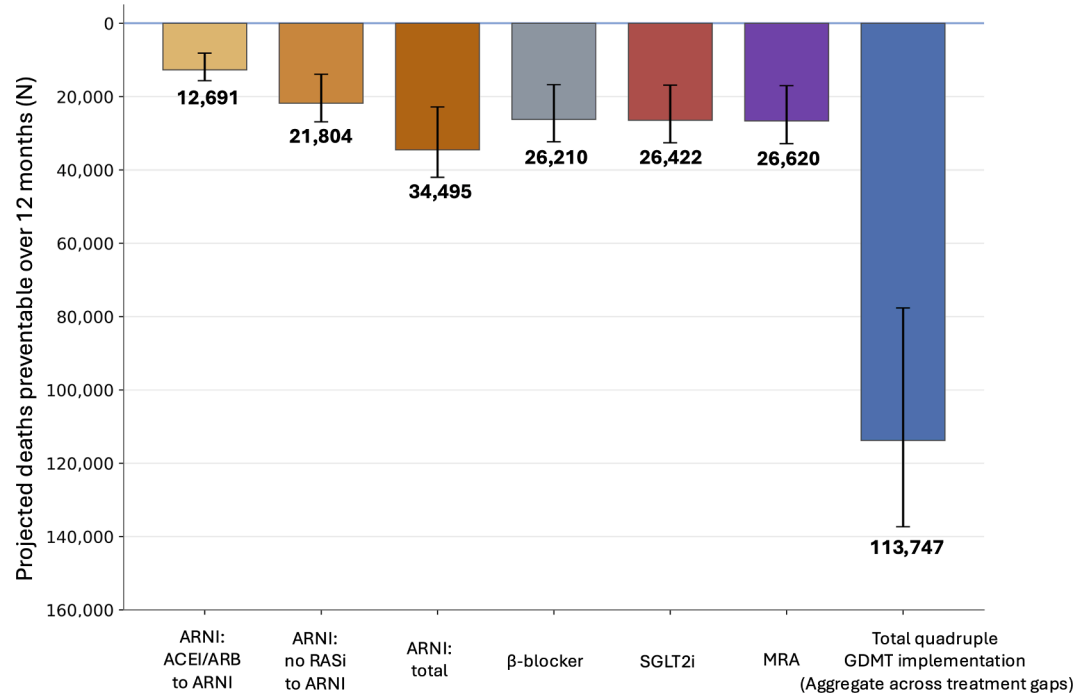


Flowchart showing sequential exclusions applied to estimate the U.S. population with chronic HFrEF and NYHA functional class II-IV symptoms for subsequent quadruple guideline-directed medical therapy modeling. HFrEF = heart failure with reduced ejection fraction; LVEF = left ventricular ejection fraction.

Sequential exclusion of patients with NYHA functional class I symptoms, hospice or comfort-care status, and receipt of advanced therapies yielded a modeled population of 2,758,648 patients with chronic symptomatic HFrEF (NYHA functional class II-IV) eligible for long-term GDMT implementation (Table 1, Figure 1).

National hospitalization data indicated 1,272,855 primary HF hospitalizations and 6,038,935 total HF hospitalizations annually. After applying the proportion attributable to HFrEF and excluding presumed new-onset HF hospitalizations, these values corresponded to 467,392 primary and 2,217,497 total HFrEF hospitalizations per year used as baseline event estimates for modeling (Table 1).

After accounting for therapy-specific contraindications, the numbers of patients eligible for individual GDMT classes remained substantial, including 2,728,303 eligible for beta-blockers, 2,532,439 for ARNIs, 2,394,507 for MRAs, and 2,584,853 for SGLT2 inhibitors (Table 1, Central Illustration). Contemporary treatment patterns demonstrated incomplete implementation across therapies, with quadruple GDMT prescribed in 18.2% of eligible patients in the primary Epic Cosmos¹² treatment-rate analysis (Table 1).

FIGURE 2 Projected Deaths Preventable Over 12 Months With Optimal Quadruple GDMT Implementation

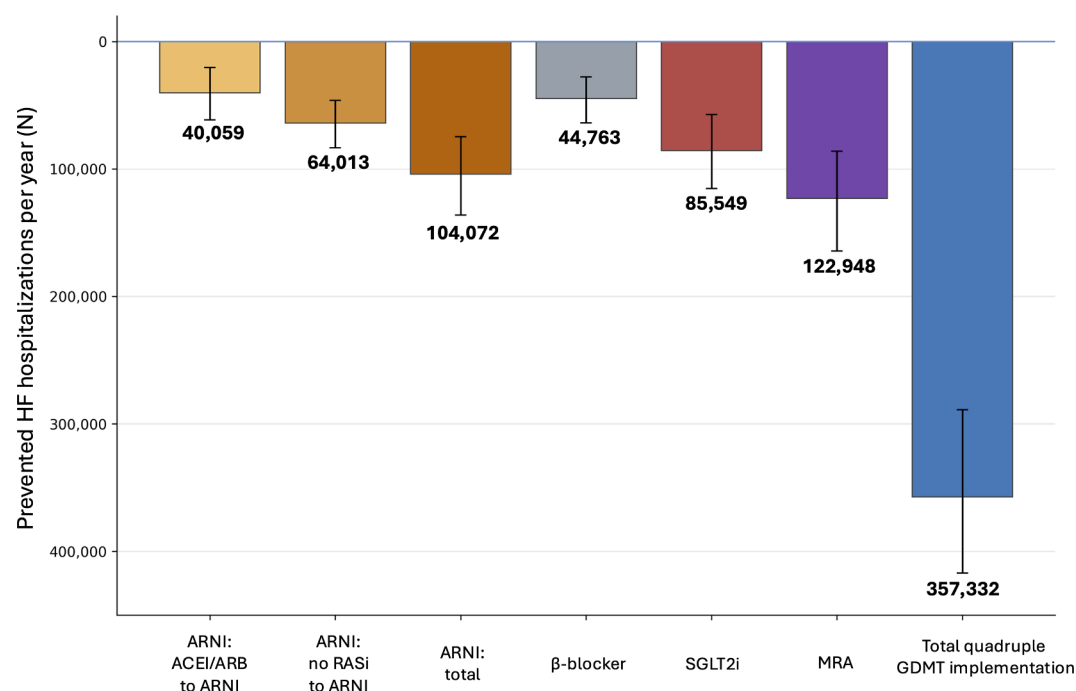
Projected deaths preventable over 12 months with optimal quadruple GDMT implementation among eligible U.S. patients with chronic HFrEF and NYHA functional class II-IV symptoms using Epic Cosmos 2023-2025 treatment rates. ARNI estimates were decomposed into patients switching from ACEI/ARB to ARNI, patients initiating ARNI from no RASi, and the combined ARNI total. The total quadruple GDMT implementation estimate represents aggregate projected deaths preventable across therapy-specific treatment gaps and should not be interpreted as a count of unique individuals. Error bars indicate 95% UIs derived using probabilistic sensitivity analysis; numbers denote point estimates. ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; GDMT = guideline-directed medical therapy; MRA = mineralocorticoid receptor antagonist; RASi = renin-angiotensin system inhibitor; SGLT2i = sodium-glucose cotransporter 2 inhibitor; UI = uncertainty interval; other abbreviation as in [Figure 1](#).

TABLE 3 Projected HF and All-Cause Hospitalizations Prevented With Optimal Quadruple GDMT Implementation

	RRR: HF Hospitalizations	Prevented HF Hospitalizations per Year	RRR: All-Cause Hospitalizations	Prevented All-Cause Hospitalizations per Year
Beta-blocker	0.360	44,763 (25,645-61,810)	0.200	117,985 (53,318-178,479)
ARNI: ACEI/ARB to ARNI	0.230	40,059 (18,704-59,655)	0.120	99,159 (37,309-158,302)
ARNI: no RASi to ARNI	0.456	64,013 (44,562-81,759)	0.261	173,831 (113,910-231,232)
ARNI: total	—	104,072 (71,988-133,569)	—	272,990 (180,192-363,267)
MRA	0.470	122,948 (81,596-159,772)	0.230	285,452 (139,417-420,809)
SGLT2 inhibitor	0.294	85,549 (55,814-113,706)	0.136	187,754 (113,149-261,904)
Total quadruple GDMT implementation	—	357,332 (297,493-425,674)	—	864,181 (671,815-1,104,143)

Values are n or n (95% UI). Projected annual HF and all-cause hospitalizations prevented with optimal quadruple GDMT implementation among U.S. patients with chronic HFrEF and NYHA functional class II-IV symptoms. HF hospitalizations were the primary hospitalization outcome; all-cause hospitalizations are shown as supportive secondary estimates. Contemporary treatment rates were derived from Epic Cosmos 2023-2025. Prevented hospitalizations were calculated by multiplying eligible untreated counts by annual baseline hospitalization rates and the corresponding hospitalization RRR. For ARNI, benefits were decomposed into patients switching from ACEI/ARB to ARNI, using PARADIGM-HF ARNI vs ACEI hospitalization treatment effects, and patients initiating ARNI from no RASi, using ARNI vs imputed placebo hospitalization treatment effects. The total quadruple GDMT implementation estimate was calculated as the sum of projected hospitalizations prevented by closing individual beta-blocker, ARNI, MRA, and SGLT2 inhibitor treatment gaps. The 95% UIs were derived using probabilistic sensitivity analysis.

RRR = relative risk reduction; other abbreviations as in [Table 1](#).

FIGURE 3 Projected HF Hospitalizations Prevented With Optimal Quadruple GDMT Implementation

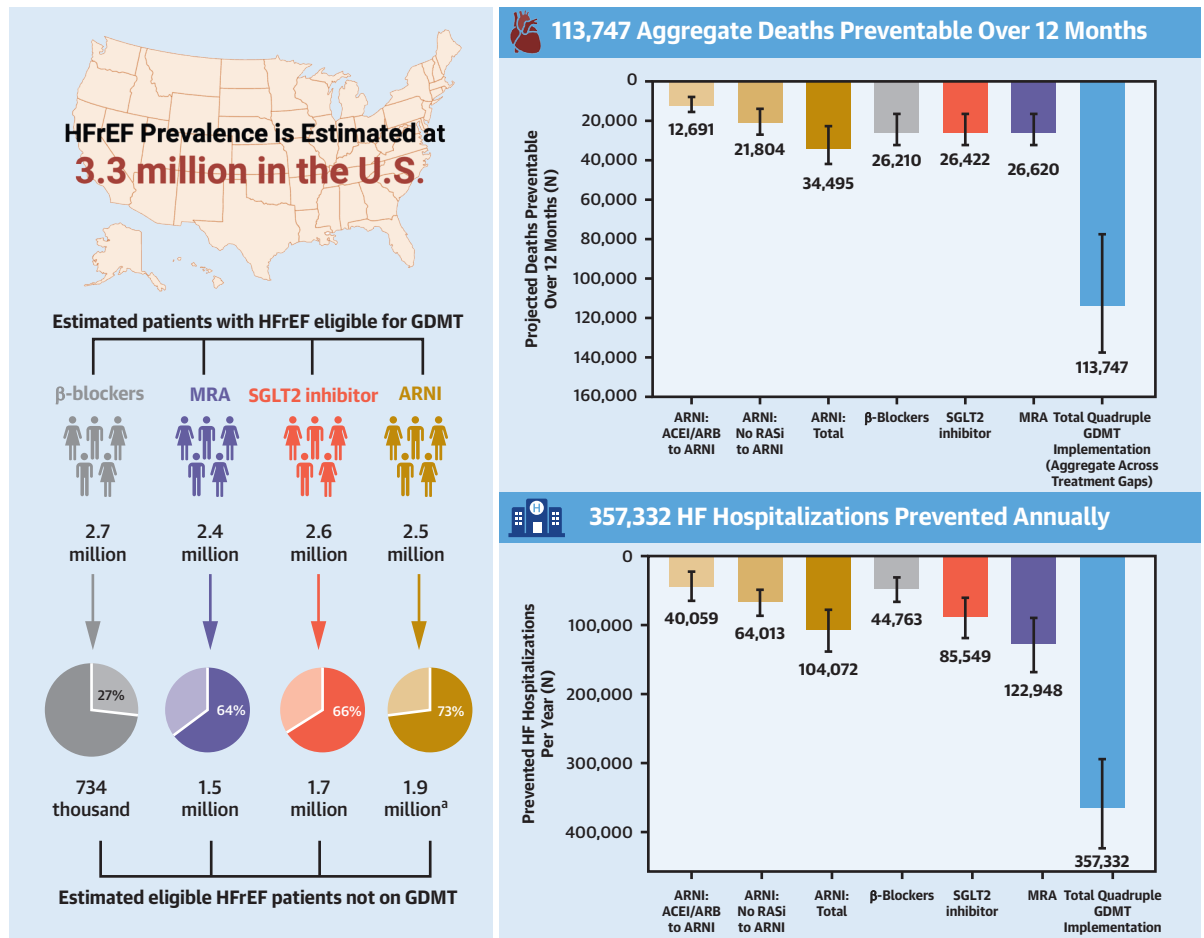
Projected annual HF hospitalizations prevented with optimal quadruple GDMT implementation among eligible U.S. patients with chronic HFrEF and NYHA functional class II-IV symptoms using Epic Cosmos 2023-2025 treatment rates. ARNI estimates were decomposed into patients switching from ACEI/ARB to ARNI, patients initiating ARNI from no RASi, and the combined ARNI total. The total quadruple GDMT implementation estimate was calculated as the sum of projected HF hospitalizations prevented by closing individual beta-blocker, ARNI, MRA, and SGLT2 inhibitor treatment gaps. Error bars indicate 95% UIs derived using probabilistic sensitivity analysis; numbers denote point estimates. HF = heart failure; other abbreviations as in [Figures 1 and 2](#).

TREATMENT GAPS AND PROJECTED DEATHS PREVENTABLE OVER 12 MONTHS WITH OPTIMAL QUADRUPLE GDMT IMPLEMENTATION. Application of contemporary EPIC COSMOS treatment rates to therapy-eligible populations revealed substantial implementation gaps across GDMT classes ([Tables 1 and 2](#)). Eligible untreated populations included 733,891 patients for beta-blockers, 1,856,531 for ARNIs, 1,543,967 for MRAs, and 1,717,444 for SGLT2 inhibitors. For ARNIs, the eligible untreated population was decomposed into 1,027,979 patients receiving ACEI/ARB but not ARNI and 828,552 patients receiving no RASi. These persistent shortfalls in adoption translated into substantial projected clinical benefit with optimal implementation.

Using trial-derived NNTs standardized to a 12-month time horizon ([Supplemental Table 2](#)), class-specific projected deaths preventable over 12 months were summed across therapy-specific treatment gaps, yielding an aggregate estimate of 113,747 deaths preventable (95% UI: 90,173-149,875) in the United States ([Table 2, Figure 2](#)). Projected deaths preventable over

12 months were 26,210 for beta-blockers, 34,495 for ARNIs, 26,620 for MRAs, and 26,422 for SGLT2 inhibitors. The ARNI estimate included 12,691 deaths preventable among patients switching from ACEI/ARB to ARNI and 21,804 among patients initiating ARNI from no RASi. The total estimate represents aggregate projected deaths preventable across therapy-specific implementation gaps and should not be interpreted as a count of unique individuals.

Findings were consistent across alternative treatment-rate sources, with projected deaths preventable over 12 months of 82,414 using GWTG-HF 2021-2024 rates, 93,546 using GWTG-HF 2016-2023 rates, and 111,845 using Optum 2020-2023 rates ([Supplemental Table 4](#)). Real-world effectiveness sensitivity analyses applying 80% and 60% of trial-derived mortality benefit yielded 90,998 and 68,248 projected deaths preventable over 12 months, respectively, in the primary Epic Cosmos analysis ([Supplemental Table 3](#)), with corresponding alternative-source analyses shown in [Supplemental Table 5](#).

CENTRAL ILLUSTRATION Projected National Impact of Optimal Implementation of Quadruple GDMT in HFrEF

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Schematic overview of the modeling framework used to estimate the population-level benefits of optimal implementation of quadruple GDMT among U.S. patients with chronic HFrEF and NYHA functional class II-IV symptoms. The illustration shows the estimated HFrEF population, therapy-specific eligibility after exclusions, current treatment gaps based on Epic Cosmos 2023-2025 treatment rates, and projected clinical benefits with optimal quadruple GDMT implementation. The mortality estimate represents aggregate projected deaths preventable over 12 months when summed across therapy-specific treatment gaps and should not be interpreted as a count of unique individuals. Annual HF hospitalizations prevented are shown as an event-based hospitalization outcome. ^aFor ARNI, the 1.9 million eligible untreated patients comprised 1,027,979 patients receiving ACEI/ARB but not ARNI and modeled as switching to ARNI using direct PARADIGM-HF treatment effects, and 828,552 patients receiving no RASi and modeled as initiating ARNI using imputed placebo-referenced treatment effects. ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; GDMT = guideline-directed medical therapy; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; MRA = mineralocorticoid receptor antagonist; PARADIGM-HF = Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure; RASi = renin-angiotensin system inhibitor; SGLT2 = sodium-glucose cotransporter 2.

PROJECTED HF HOSPITALIZATIONS PREVENTED WITH OPTIMAL QUADRUPLE GDMT IMPLEMENTATION. Trial-derived hospitalization treatment effects used for class-specific modeling are summarized in [Supplemental Table 6](#). Application of these effects to national hospitalization baselines demonstrated

substantial reductions in annual HF hospitalizations with optimal quadruple GDMT implementation ([Table 3](#), [Figure 3](#)). Across individual GDMT classes, projected annual HF hospitalizations prevented were 44,763 for beta-blockers, 104,072 for ARNIs, 122,948 for MRAs, and 85,549 for SGLT2 inhibitors. For ARNI,

projected HF hospitalizations prevented included 40,059 among patients switching from ACEI/ARB to ARNI and 64,013 among patients initiating ARNI from no RASi.

Overall, total quadruple GDMT implementation was projected to prevent 357,332 HF hospitalizations annually (95% UI: 297,493-425,674) (**Table 3**, **Figure 3**). All-cause hospitalization estimates were retained as supportive secondary results in **Table 3** and **Supplemental Figure 1**. Total quadruple GDMT implementation was projected to prevent 864,181 all-cause hospitalizations annually (95% UI: 671,815-1,104,143).

Across alternative treatment-rate sources, total quadruple GDMT implementation was projected to prevent 274,215 HF hospitalizations annually using GWTG-HF 2021-2024 rates, 318,795 using GWTG-HF 2016-2023 rates, and 366,915 using Optum 2020-2023 rates (**Supplemental Table 8**). Real-world effectiveness sensitivity analyses applying 80% and 60% of trial-derived hospitalization benefit yielded 285,865 and 214,399 projected HF hospitalizations prevented annually, respectively, in the primary Epic Cosmos analysis (**Supplemental Table 7**), with corresponding alternative-source analyses shown in **Supplemental Table 9**.

DISCUSSION

In this population-level decision analytic modeling study using contemporary U.S. epidemiologic data, we estimated the potential population health impact of optimal quadruple GDMT implementation among patients with chronic symptomatic HFrEF. After applying standard clinical exclusions and contraindication adjustments, approximately 2.76 million U.S. patients were eligible for foundational GDMT, yet contemporary Epic Cosmos treatment rates demonstrated substantial implementation gaps, with only 18.2% receiving quadruple therapy. Optimal implementation among eligible untreated patients was projected to prevent a cumulative total of 113,747 deaths over 12 months when summed across the individual therapy-specific treatment gaps; because a given patient may be untreated for >1 therapy, this total reflects aggregate preventable deaths across treatment gaps rather than a count of unique individuals. Optimal implementation was also projected to prevent 357,332 HF hospitalizations annually. These findings provide updated national benchmarks for evaluating the population-level benefits of improving GDMT adoption and demonstrate that incomplete implementation of evidence-based therapies remains one of the most

consequential targets for reducing morbidity and mortality in contemporary HF care.

The scale of these projections establishes GDMT implementation as a major public health priority and is consistent with recent global modeling by Tang et al,²⁷ which projected that optimal GDMT implementation could prevent 1.19 million deaths worldwide per year, with ARNIs and SGLT2 inhibitors accounting for the largest proportions of projected mortality benefit. Together, these analyses support a coherent international picture: GDMT underutilization represents a preventable mortality burden, with direct implications for evidence-based policymaking and appraisal of GDMT optimization relative to other competing clinical and public health investments.

The primary estimates rest on evidence that GDMT class benefits are incremental when therapies are combined. Vaduganathan et al²⁵ demonstrated that escalation to full quadruple therapy was associated with substantial reductions in all-cause mortality and cardiovascular death or HF hospitalization compared with ACEI/ARB plus beta-blocker alone, whereas Tromp et al²⁶ found that quadruple therapy produced the greatest estimated benefit across all pharmacologic combinations. Critically, these analyses suggest that partial adoption, the predominant real-world pattern, forfeits a disproportionate share of the achievable benefit, with consequences falling most heavily on patients least likely to receive complete therapy.

The primary estimates represent an optimal-implementation scenario, assuming appropriate initiation, dosing, adherence, persistence, and monitoring. Available real-world data generally support the effectiveness of GDMT in broader clinical populations, although realized benefits may be attenuated when adherence, dose optimization, follow-up intensity, affordability, or comorbidity burden differ from clinical trial conditions.^{28,29} Importantly, even under conservative assumptions, the opportunity remains large: the 80% and 60% real-world effectiveness sensitivity analyses yielded aggregate estimates across therapy-specific treatment gaps of 90,998 and 68,248 deaths preventable over 12 months, respectively, suggesting that substantial gains are achievable well short of optimal trial-equivalent implementation.

The findings of this analysis also have direct implications for health care resource allocation. All 4 pillars of GDMT for HFrEF are cost-effective under conventional U.S. thresholds. Generic beta-blockers, ACEI/ARBs, and MRAs have incremental cost-effectiveness ratios below \$1,000/quality adjusted life years, whereas sacubitril/valsartan and SGLT2

inhibitors also meet conventional cost-effectiveness thresholds.^{2,30} Importantly, sacubitril/valsartan is now available in generic formulation, and dapagliflozin is now available as a lower-cost generic SGLT2 inhibitor option. These developments are expected to reduce the medication-cost burden of quadruple GDMT, further lowering affordability barriers and strengthening the rationale for timely implementation.^{31,32}

Despite compelling randomized trial evidence, real-world GDMT implementation remains incomplete because of multilevel barriers spanning patients, clinicians, and health systems.^{33,34} Clinical inertia, underappreciation of patient risk, misconstruing stable symptoms as low risk, concerns about renal dysfunction or hyperkalemia, medication affordability, polypharmacy, limited health literacy, fragmented care, restrictive coverage policies, medication costs, and structural inequities all contribute to undertreatment and poor adherence.³⁵⁻³⁷ The risk-treatment paradox further compounds these gaps, with older adults, patients with chronic kidney disease, those with more severe symptoms, and socially or economically marginalized populations often less likely to receive appropriate therapies despite substantial potential benefit.³⁴

Several evidence-based mechanisms exist to improve GDMT implementation. The STRONG-HF (Safety, Tolerability and Efficacy of Rapid Optimization, Helped by NT-proBNP Testing, of Heart Failure Therapies) trial demonstrated that high-intensity management with rapid GDMT up-titration and close postdischarge follow-up was safe, well tolerated, and associated with a significant reduction in 180-day all-cause death or HF readmission compared with usual care.³⁸ At the health system level, electronic health record-based clinical decision support algorithms that identify eligible untreated patients and prompt optimization have shown promise in improving initiation rates, and integration of pharmacists, patient navigators, and registered nurses in collaborative practice models has been associated with improved titration and adherence. At the policy level, lower prescription insurance copays and value-based insurance designs that align cost-sharing with therapeutic value have been associated with improved adherence to cardiovascular medications.³⁹ Finally, implementation science approaches specifically designed to address inequities in GDMT adoption, among underrepresented racial and ethnic groups, women, patients in lower socioeconomic strata, and those in rural regions, should be treated as a research and policy priority in their own right, not as downstream considerations.

STUDY LIMITATIONS. First, this model used aggregate data from published studies, registries, and electronic health record datasets without patient-level treatment profiles. Thus, overlap in untreated eligibility across GDMT classes could not be directly modeled, and the total projected deaths preventable should be interpreted as aggregate projected deaths preventable across therapy-specific gaps rather than as a count of unique individuals. Second, the model assumes incremental benefits across GDMT classes; although supported by previous comparative and network meta-analytic studies,^{25,26} true combined effects may differ in routine practice. Third, ARNI benefits for patients initiating therapy from no RASi were indirectly estimated from trials conducted in different eras and populations, introducing uncertainty despite separately modeling ACEI/ARB-to-ARNI switching using the direct PARADIGM-HF comparison. Fourth, standardizing NNTs to 12 months assumes approximately constant hazards and linear accumulation of benefit over time. Fifth, primary estimates represent optimal implementation, including appropriate initiation, dosing, adherence, persistence, and monitoring; realized benefits may be lower in practice, as explored in 80% and 60% real-world effectiveness sensitivity analyses. Sixth, contraindication rates, treatment rates, and the 20% new-onset HF hospitalization exclusion were derived from available literature and real-world data sources and may not fully represent all U.S. practice settings. Finally, hospitalization estimates used average baseline event rates and administrative data; all-cause hospitalization estimates are less directly attributable to GDMT than HF hospitalizations and should be interpreted as supportive secondary estimates.⁴⁰

CONCLUSIONS

This population-level modeling study demonstrates that incomplete implementation of GDMT in HFrEF represents one of the largest modifiable opportunities to reduce cardiovascular morbidity and mortality in the United States. Optimal implementation of quadruple therapy among eligible patients was associated with an aggregate estimate exceeding 113,000 deaths preventable over 12 months across therapy-specific treatment gaps and >357,000 HF hospitalizations prevented annually, a magnitude that places the GDMT implementation gap among the most consequential addressable targets in contemporary public health. These updated national estimates provide benchmarks for evidence-based policymaking, health care system performance evaluation, and

prioritization of implementation investments. Closing this gap will require coordinated action across clinician practice, health system design, and coverage policy, yet the data presented here suggest that few interventions in cardiovascular medicine offer a comparable return on that investment.

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PERSPECTIVES

COMPETENCY IN SYSTEMS-BASED PRACTICE: Among patients in the U.S. with chronic symptomatic HFrEF, incomplete implementation of quadruple GDMT represents a major modifiable care gap, with optimal implementation projected to prevent 113,747 deaths over 12 months and 357,332 HF hospitalizations annually.

TRANSLATIONAL OUTLOOK: Future implementation studies should test scalable, health-system-level strategies to identify eligible untreated patients, optimize quadruple GDMT initiation and persistence, and determine whether projected population-level reductions in mortality and HF hospitalizations can be achieved in routine practice.

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APPENDIX For an expanded Methods section as well as supplemental tables and figure, please see the online version of this paper.